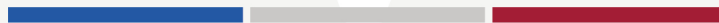




MIT SLOAN SCHOOL OF MANAGEMENT
MIT SCHWARZMAN COLLEGE OF COMPUTING



MODULE 3 UNIT 1
Lesson Video 3 Transcript

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Structural shift

PAUL MCDONAGH-SMITH: We've examined the paradox of scaling and the practical architecture of trust. Now I want to zoom out because scaling agentic AI isn't only an operational challenge. It's also a strategic one. And the strategic landscape is shifting beneath us in ways that have implications well beyond any single agent or agentic workflow. Let me frame what I mean.

The economics of software are changing

MCDONAGH-SMITH: For the past two decades, the dominant model of enterprise technology has been buy, configure, subscribe. Organizations purchase package software, CRM systems, ERP platforms, analytics suites, because the cost of building that functionality internally was prohibitive. You paid the license fee because writing the code yourself was slower, more expensive, and harder to maintain. Agentic AI is potentially disrupting that equation.

The cost of producing software is falling dramatically. Coding assistance and agent-based development tools are enabling smaller teams to build in days or weeks. What previously required months and large engineering organizations. And this isn't just about prototyping. These systems are increasingly capable of generating production-quality code, testing it, and even participating in its ongoing maintenance. The implication for your scaling strategy is potentially profound. The build versus buy calculus for enterprise software is being rewritten in real time.

Functions that you currently pay external vendors to perform may become viable to build internally. Not because your engineering team has grown, but because the tools they work with have fundamentally changed what a small group of people can accomplish. This doesn't mean every organization should immediately build everything in-house, but it does mean that the assumptions underpinning your technology strategy, what is commodity, what is differentiating, and what is worth paying a premium for, needs to be re-examined more frequently than they probably have been. The half-life of a sound technology architecture decision is getting shorter.

The emerging architecture: Intent over code

MCDONAGH-SMITH: There's a shift underway that I want to put onto your radar, even if it hasn't yet fully arrived, and it concerns the nature of software itself. Historically, if you wanted a computer system to do something, you wrote code, precise, syntactically exact instructions in a formal programming language. Each generation of programming language has raised the level of abstraction from machine code to assembly to high-level languages like Java and Python to declarative languages like sequel, SQL, where you specify what you want and let the system figure out how to deliver it. The trajectory points towards a further leap.

Systems where the specification is written not in a programming language at all, but in structured natural language. You describe in detailed rigorous language what the system should do, how it should behave, what its constraints are, and agents generate the code,

deploy it, and maintain it. In this model, the source code is no longer the intellectual asset. The intellectual asset is the specification, what some researchers call the system ontology. A structured description of your business logic, your rules, your workflows, your intent. Code becomes a derivative, automatically generated and continuously regenerated as the underlying models improve.

Organizations would not buy software; they would buy and customize best practice ontologies, structured descriptions of how a business function could, should operate, and agents would build the software from those descriptions. The direction of travel is clear and the pace is accelerating. For leaders thinking about scaling strategy, the question this raises is, “Are you investing?” Are you investing in assets that will appreciate in an AI-native world? Structured knowledge, clean data, well-defined processes, or in assets that may depreciate, rigid custom code, tightly coupled vendor dependencies, manual processes that resist redesign.

Organizational readiness: The binding constraint

MCDONAGH-SMITH: And this brings us to what I consider an important and often underestimated dimension of scaling: organizational readiness. The technology will keep improving. The models will get better. The tools will get cheaper. Those trends aren't in question. What is in question is whether your organization, your people, your culture, your incentive structures can absorb and direct those capabilities at the pace they are arriving.

Scaling agentic AI touches identity. For a professional who spent perhaps decades building expertise in a particular domain, writing code in a specific language, managing a specific process, holding institutional knowledge that others depend upon, the arrival of systems that can perform aspects of that work isn't a neutral event. It's an identity question. And leaders who ignore the human dimension of that transition will find that their most capable people become their most significant source of resistance, potentially. The organizations that are scaling most effectively tend to share two characteristics.

First, they pursue a dual strategy, top-down sponsorship that creates permission and resources, combined with bottom-up experimentation that lets individuals and small teams discover value for themselves. The top-down provides direction, the bottom-up provides conviction. You need both.

Second, they compress experimentation cycles ruthlessly. Rather than large planning committees and extended evaluation periods, they form small, focused teams. Three, four, five people freed from their normal meeting loads with a clear mandate: build something real in a matter of weeks. The results from these teams tend to be disproportionately impressive, not because the people in themselves are exceptional, but because the constraints are liberating. Small scope, short timelines, fast feedback.

Conclusion

MCDONAGH-SMITH: So here's the strategic picture. Scaling agentic AI isn't just about deploying more agents to more tasks. It's about positioning your organization at the intersection of three shifts. The falling cost of software, the rising power of agent-based workflows, and the urgent need for a workforce that can adapt, experiment, and lead in a landscape that will not sit still.

In the sessions ahead, we'll explore this frontier, and you'll continue building your own strategic playbook, integrating the risk, the governance, and the scaling considerations we've been developing throughout this program. The opportunity is real, but it favors the prepared.

SPEAKER: Did you understand all the concepts covered in this video? If you'd like to review any of the content, please click on the chapter menu.